

A Strong Operator Riccati Solution whose Graph Is Not Reducing

Automated mathematics run `fa_banach_001`, lane 1

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Status. Candidate counterexample, likely valid, pending expert review. Under the exact hypotheses of Makarov–Schmitz–Seelmann, strong solvability of the operator Riccati equation does not imply that the associated graph subspace is reducing.

1 Source problem

Makarov, Schmitz, and Seelmann [1] consider

$$A = \begin{pmatrix} A_0 & 0 \\ 0 & A_1 \end{pmatrix}, \quad V = \begin{pmatrix} 0 & W \\ W^* & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -X^* \\ X & 0 \end{pmatrix},$$

where A_0, A_1 are self-adjoint, X is bounded, $\text{Dom}(W) \supset \text{Dom}(A_1)$, and $\text{Dom}(W^*) \supset \text{Dom}(A_0)$. They compare

- (i) $\mathcal{G}(\mathcal{H}_0, X)$ is reducing for $A + V$;
- (ii) Y is a strong solution of

$$AY - YA - YVY + V = 0. \tag{1}$$

On page 3 they state that they have no counterexample in which only one of these requirements holds, and ask whether the two are equivalent or logically independent.

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where U is the unitary transformation from the polar decomposition $I_{\mathcal{H}} + Y = U|I_{\mathcal{H}} + Y|$ (hence $I_{\mathcal{H}} - Y = |I_{\mathcal{H}} + Y|U^*$), and the diagonal entries B_0 and B_1 are similar to the corresponding diagonal entries of the right-hand sides of (1.4) and (1.5) (see Remark 2.11).

Although we are not able to prove that (i) and (ii) are equivalent in general, we do not have an appropriate counterexample for which only one of the two requirements holds. Hence, in the framework of our approach, it remains an open problem whether the requirements (i) and (ii) are (logically) independent or not.

However, under natural additional regularity conditions that automatically hold if, for instance, the off-diagonal perturbation V is bounded, we show that the properties (i) and (ii) are indeed equivalent. In this case, the operator $A + V$ can be diagonalized in either of the two forms

Figure 1: The open problem in the introduction of [1, p. 3].

The associated graph is $\mathcal{G}(\mathcal{H}_0, X) = \{f \oplus Xf : f \in \mathcal{H}_0\}$. “Reducing” includes both invariance of this graph and its orthogonal complement and the domain splitting by their intersections with $\text{Dom}(A + V)$. “Strong” means that $Y \text{Dom}(A) \subset \text{Dom}(A)$ and (1) holds throughout $\text{Dom}(A)$.

2 Proof intuition

Domain preservation by a bounded operator need not pass to its bounded inverse when the domain carries a much stronger graph norm. We exploit that gap with a Toeplitz operator X which has Hilbert-space norm at most $1/20$ but whose realization on a rapidly weighted domain has i in its spectrum. Thus X preserves the domain while $(I + X^2)^{-1}$ does not. The first fact is exactly the domain part of strong Riccati solvability; the second obstructs the domain splitting required for a reducing graph.

It remains to choose the off-diagonal operator. Writing $C = AX - XA$, the Riccati equation reduces to $W + XWX = C$. Because X is tiny in the Hilbert norm even though it is large in the graph norm, the alternating series $\sum_{n \geq 0} (-1)^n X^n C X^n$ converges on the weighted domain, telescopes to the required identity, and is skew-symmetric. This supplies W and W^* with exactly the domains required by the source hypotheses.

3 Counterexample theorem

Theorem 1. *There are separable Hilbert spaces $\mathcal{H}_0 = \mathcal{H}_1$, self-adjoint operators $A_0 = A_1$, a densely defined operator W satisfying*

$$\text{Dom}(W) \supset \text{Dom}(A_1), \quad \text{Dom}(W^*) \supset \text{Dom}(A_0),$$

and a bounded positive self-adjoint $X : \mathcal{H}_0 \rightarrow \mathcal{H}_1$ such that the operator Y above is a strong solution of (1), but $\mathcal{G}(\mathcal{H}_0, X)$ is not reducing for $A + V$.

Consequently requirement (ii) in the source problem does not imply requirement (i), and the two requirements are not equivalent in the generality of Hypotheses 2.1–2.2 of [1].

Proof. Let $K = \ell^2(\mathbb{N}_0)$, with standard basis $(e_n)_{n \geq 0}$, and let $Se_n = e_{n+1}$ be the unilateral shift. Define

$$Ae_n = 200^n e_n, \quad \mathcal{D} := \text{Dom}(A) = \left\{ u = (u_n) : \sum_{n \geq 0} 200^{2n} |u_n|^2 < \infty \right\}.$$

Then A is self-adjoint, $A \geq I$, and $A : \mathcal{D} \rightarrow K$ is an isometric isomorphism when $\mathcal{D}$ is given the norm $\|u\|_{\mathcal{D}} = \|Au\|$.

Put

$$X = \frac{3}{100}I + \frac{1}{100}(S + S^*). \tag{2}$$

Since $-2I \leq S + S^* \leq 2I$, the operator X is positive and

$$\|X\| \leq \frac{1}{20} =: q. \tag{3}$$

Both S and S^* preserve $\mathcal{D}$, so $X\mathcal{D} \subset \mathcal{D}$. Conjugating its restriction to the graph-norm space gives the bounded operator

$$Z := AXA^{-1} = \frac{3}{100}I + 2S + \frac{1}{20000}S^*, \quad \|Z\| \leq \frac{40601}{20000} =: M. \tag{4}$$

We first prove the crucial failure

$$(I + X^2)^{-1}\mathcal{D} \not\subset \mathcal{D}. \tag{5}$$

Indeed, consider the quadratic polynomial

$$p(z) = 2z^2 + \left(\frac{3}{100} + i\right)z + \frac{1}{20000}.$$

On $|z| = 1$,

$$\left|\left(\frac{3}{100} + i\right)z + \frac{1}{20000}\right| \leq \sqrt{1 + \frac{9}{10000}} + \frac{1}{20000} < 2 = |2z^2|.$$

Rouché's theorem puts both roots of p in the open unit disk. Set $v_{-1} = 0$, $v_0 = 1$, and recursively

$$2v_{n+1} + \left(\frac{3}{100} + i\right)v_n + \frac{1}{20000}v_{n-1} = 0.$$

Since both characteristic roots are in the open disk, the nonzero sequence $v = (v_n)$ belongs to $\ell^2(\mathbb{N}_0)$. Formula (4) gives $(Z^* + i)v = 0$, whence $i \in \text{spec}(Z)$. By polynomial spectral mapping, $I + Z^2$ is not invertible.

If $R := (I + X^2)^{-1}$ mapped $\mathcal{D}$ into itself, then its restriction on the Banach space $\mathcal{D}$ would be bounded by the closed graph theorem: convergence in $\mathcal{D}$ implies convergence in K , where R is bounded. Hence ARA^{-1} would be a bounded two-sided inverse of $I + Z^2$, because $AX^2A^{-1} = Z^2$. This contradiction proves (5).

Now define on $\mathcal{D}$

$$C := AX - XA, \quad W := \sum_{n=0}^{\infty} (-1)^n X^n C X^n.$$

This series converges in K for every $u \in \mathcal{D}$. In fact,

$$\begin{aligned} \|X^n C X^n u\| &\leq q^n (\|AX^{n+1}u\| + q\|AX^n u\|) \\ &\leq (M + q)(qM)^n \|Au\|, \end{aligned}$$

and

$$qM = \frac{40601}{400000} < 1.$$

Thus $W : \mathcal{D} \rightarrow K$ is well-defined and graph-norm bounded. The operator C is skew-symmetric on $\mathcal{D}$, and so is every $X^n C X^n$; absolute convergence of the scalar products shows that W is skew-symmetric. In particular W is densely defined and

$$\mathcal{D} \subset \text{Dom}(W^*), \quad W^*u = -Wu \quad (u \in \mathcal{D}). \quad (6)$$

Moreover, adding the two convergent series and shifting the index yields the telescoping identity

$$W + XWX = C = AX - XA \quad \text{on } \mathcal{D}. \quad (7)$$

Take $\mathcal{H}_0 = \mathcal{H}_1 = K$, $A_0 = A_1 = A$, and $\text{Dom}(W) = \mathcal{D}$, and form

$$\mathbf{A} = \begin{pmatrix} A & 0 \\ 0 & A \end{pmatrix}, \quad V = \begin{pmatrix} 0 & W \\ W^* & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -X \\ X & 0 \end{pmatrix}.$$

The domain hypotheses on W follow from (6). Since $X\mathcal{D} \subset \mathcal{D}$, the operator Y maps $\text{Dom}(\mathbf{A}) = \mathcal{D} \oplus \mathcal{D}$ into itself. The two off-diagonal block equations in (1) both reduce, using (6), to

$$AX - XA - XWX - W = 0,$$

which is exactly (7). Hence Y is a strong Riccati solution.

It remains to disprove reduction. The Riccati identity already makes $\mathcal{G}(K, X)$ and $\mathcal{G}(K, -X)$ invariant for $\mathbf{A} + V$. Their domain intersections sum to

$$T(\mathcal{D} \oplus \mathcal{D}), \quad T = I + Y = \begin{pmatrix} I & -X \\ X & I \end{pmatrix}.$$

Because $X = X^*$,

$$T^{-1} = \begin{pmatrix} R & XR \\ -XR & R \end{pmatrix}, \quad R = (I + X^2)^{-1}.$$

By (5), some $f \in \mathcal{D}$ has $Rf \notin \mathcal{D}$. Therefore $T^{-1}(f \oplus 0) \notin \mathcal{D} \oplus \mathcal{D}$, so $T(\mathcal{D} \oplus \mathcal{D}) \neq \mathcal{D} \oplus \mathcal{D}$. The domain does not split along the two invariant graphs, and $\mathcal{G}(K, X)$ is not reducing. $\square$

4 Verification, scope, and novelty

The proof is exact. The deterministic script `code/check_parameters.py` verifies the rational contraction and geometric-series bounds, the strict Rouché inequality, and the two characteristic roots. It is an illustration rather than a substitute for the operator proof. A separate sign-by-sign block multiplication and the domain-transfer argument are recorded in `VERIFICATION.md`.

This example completely disproves the implication “strong solution $\Rightarrow$ reducing graph” and therefore disproves general equivalence. It does not disprove the converse implication. Thus, if “logical independence” is reserved for failure of both implications, one half of that stronger statement remains open.

The local run indexes were searched for arXiv:1307.6439, the exact title, and the core Riccati/graph terms. Bounded web searches on August 9, 2026 used the exact open-problem sentence, the exact title and authors, and combinations of “strong solution,” “reducing graph,” and “counterexample.” They found the source paper and the authors’ later paper [2], which still calls the corresponding domain-splitting/domain-invariance question open, but no later resolution or this construction. Novelty confidence is positive but conservative pending expert literature review.

Human review is strongly recommended. The decisive checks are the passage from $i \in \text{spec}(Z)$ to failure of domain preservation, convergence and skew symmetry of the series defining W , and the signs in the two Riccati blocks.

References

- [1] K. A. Makarov, S. Schmitz, and A. Seelmann, *Reducing graph subspaces and strong solutions to operator Riccati equations*, unpublished preprint, arXiv:1307.6439 (2013).
- [2] K. A. Makarov, S. Schmitz, and A. Seelmann, *On invariant graph subspaces*, Integral Equations Operator Theory **85** (2016), 399–425; arXiv:1509.07984.